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Review

Review on recent Photovoltaic/Thermal (PV/T) technology advances and applications

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ABSTRACT

This paper shows a review of the recent literature on Photovoltaic/Thermal PV/T technology and application that is carried out by various researchers and scientists. It describes various types of PV/T collectors such as air, water/air and water PV/T collectors in terms of performance, design, fabrication, simulation and experimental evaluation. Also, it evaluates the applications of PV/T technology such as building integrated photovoltaic/thermal (BIPVT) collector, Photovoltaic-Thermal/heat pump systems, water desalination, solar still, solar cooling and solar greenhouse. Results of this paper showed that energy production of PVT systems has been increased significantly. From the current literature reviews, PV/T system is a very promising technology and needs a further development in term of the overall efficiency and cost reduction.

1. Introduction

Nowadays, an extreme importance has been placed on the rational use of energy sources (Hawlader and Jahangeer, 2006). The undesirable impacts of the conventional energy resources on the environment and the depletion of some of these resources have created a major interest for the use of non-polluting, non-depletable and site dependent energy resources such as solar energy (Elhadidy and Shaahid, 2000). Studies have shown that the use of solar thermal is only 0.5% of the total primary energy need and solar photovoltaic has even lower energy supply ratio that is 0.04% (IEA/OECD, 2007). Both solar thermal and PV technologies have far high space to improve which would be driven by the continuous technical advances and increased concerns of energy saving and environment protection. This development would certainly contribute to significant reduction of fossil fuel consumption and cut of carbon emission.

Solar thermal and PV technologies can be combined together to form a single module called a Photovoltaic/Thermal system (PV/T). The PV/T is a hybrid technology consisting of solar photovoltaic cells (PV) and solar thermal components into a single module to generate both electricity and heat and improve the efficiency of solar conversion (Zhang et al., 2012).

The PV/T module has many applications. It can be used in the domestic and in the industrial field, for the purpose of preheating air or water. PVT/air systems can be integrated on building facades and roofs and apart of the electrical load they can provide building air ventilation and heating needs. On the other hand, PVT/water systems consist of a water heat exchanger that is in thermal contact with the photovoltaic back side and are suitable for space heating and water heating, etc. (Tripanagnostopoulos, 2007).

2. Types of PV/T collectors

2.1. PV/T air type collectors

Air acts as a working fluid in PV/T system. It's used to keep the temperature of PV module at optimum levels, thus high production of power can be obtained. Moreover, the heating system is used for drying, space heating and ventilation, etc.

2.1.1. Flat plate air collector

Flat plate collector is considered the main part of any solar system. It is used to absorb and collect energy from the sun and convert it into heat, then transfer the heat to stream of liquid or gases (Tyagi et al., 2012).

i. Performance and design evaluation

The performance of the air flow rate for single pass, double duct with fins PV/T collectors was elaborated (Alfegi et al., 2009). The collector has three major components: the glass cover, photovoltaic cells and the absorber plate. To enlarge the ability of removing heat

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from photovoltaic cells, aluminum fins are attached to the collector. An experiment was conducted to evaluate the effects of the mass flow rate on the efficiency of the system. Investigations showed that the increment in the mass flow rate will yield to increase in the electrical efficiency.

Tiwari and Sodha (2007) investigated the performance of PV/T air collector. A glazed PV/T air collector, with and without tedlar design configuration was considered. Besides, they derived the analytical expression of the temperature PV cells, back surface of the module, the extraction rate of the useful thermal energy of the system. Results showed that the glazed PV/T collector without tedlar provides the best performance. The combined efficiencies of the system (thermal + electrical) is increased with increase of the air mass flow rate due to reduction in the system losses, and decreases with increase of module's length and the photovoltaic cell temperature.

Tonui and Tripanagnostopoulos (2008) presented natural air flow PV/T systems. They studied the enhancements of heat transfer in the air channel through suspending of metal sheets and/or fins in the middle and the back of the air channel. Besides, a numerical model was developed and validated the experimental data. They concluded that there is a close agreement between the measured and predicted data.

Kumar and Rosen (2011) developed a double pass PV/T air collector with vertical fins in the lower air channel (Fig. 1). They founded that fins with large area can significantly reduces cell temperature from 88 °C to 66 °C. Also, the higher the packing factor the more production of the electrical output per unit collector area, this will yield in

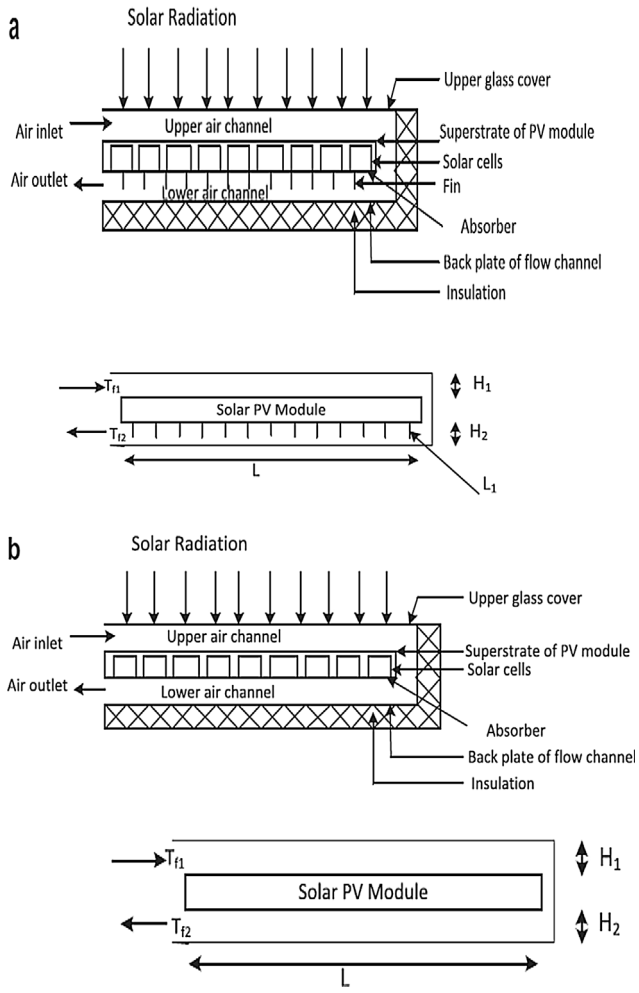


Fig. 1. (a) Cross-sectional view of double-pass PV/T solar air heater with fins. (b) Cross-sectional view of double-pass PV/T solar air heater without fins (Kumar and Rosen, 2011).

controlling the cell temperature.

Bambrook and Sproul (2012) performed a graphical representation of rate of heat output and the temperature rise as a function of the number of transfer units NTUs whereby the influence of fundamental parameters values on the thermal performance of the PV/'T air collector was illustrated (Fig. 2). Aiming to maximize the thermal and electrical output, an unglazed, single pass, open loop PV/T air system was designed in Sydney. The PV/T air collector was positioned towards the north with a tilt angle of 34°and used six 110 Wp frameless PV modules (Fig. 3). They concluded that the thermal (12–55%) and electrical (10.6–12.2%) efficiencies can be increased by increasing the air mass flow rate.

A detailed investigation was carried out to study the performance of a solar photovoltaic thermal (PV/T) air collector (Sarhaddi et al., 2010). An electrical and thermal model were developed to calculate the electrical and thermal parameters of a typical PV/T air collector. These parameters are outlet air temperature, back surface temperature, solar cell temperature, etc. In order to improve the thermal model of PV/T air collector, some corrections were made on heat loss coefficients. Besides, development on the electrical model was carried out to estimate the electrical parameters such as open-circuit voltage, short-circuit current, maximum power point voltage and current. A great agreement achieved between results of numerical simulation and the experimental measurements. It was found that the electrical, thermal and the combined system efficiency is 17.18%, 10.01% and 45%, respectively.

Investigations on the Exergetic performance assessment of a photovoltaic thermal (PV/T) air collector were studied (Sarhaddi et al., 2010). Exeregy and energy analysis were evaluated to calculate the electrical, thermal, energy and exeregy efficiencies. Development of simulation program was produced to calculate the system output efficiency (electrical/thermal). It was concluded that the simulation results are in good agreement with the experimental data.

The performance of hybrid photovoltaic thermal double pass facade for space heating was evaluated (Kamthania et al., 2011). By using the energy balance equations, the thermal modeling was developed under quasi-steady state conditions. Numerical expressions have been analyzed. It was showed that the annual electrical and thermal energy are 469.87 kWh and 480.81 kWh respectively.

The performance evaluation of single pass photovoltaic/thermal air collector with rectangular tunnel absorber has been carried out (Jin et al., 2010). An experiment has been conducted to investigate a reasonable air flow rate for heat extraction from PV cells (Fig. 3, and thus maximizing the electrical efficiency. Results revealed that the use of rectangular tunnel absorber gives better performance.

Othman et al. (2007) studied the performance of double pass PV/T collectors with fins. Fig. 4; shows the hybrid system that is consisting of PV panels and fins pasted on and under the absorber plate, respectively. A mathematical model was developed; they concluded that the proposed hybrid system can maximize the total output efficiency.

2.1.1.1. Analytical modeling. Dubey et al. (2009) derived the analytical expressions for N hybrid photovoltaic/thermal (PV/T) air collectors were connected in series (Fig. 5). Analysis of energy, exergy and electrical energy were evaluated by varying the number of collectors and air velocity. They concluded that collectors fully covered by PV module and air flows located under absorber plate give better performance in terms of thermal energy, electrical energy and exergy gain.

2.1.1.2. Experimental work. Shahsavar and Ameri (2010) conducted an experimental investigation and modeling of a direct-coupled PV/T air collector in Iran. The system has an aluminum sheet located at the middle of the air channel to improve the extraction of heat from photovoltaic panels. They developed a theoretical model to compare it with the experimental data. Results revealed that there was a good agreement between the simulation data and the experimental

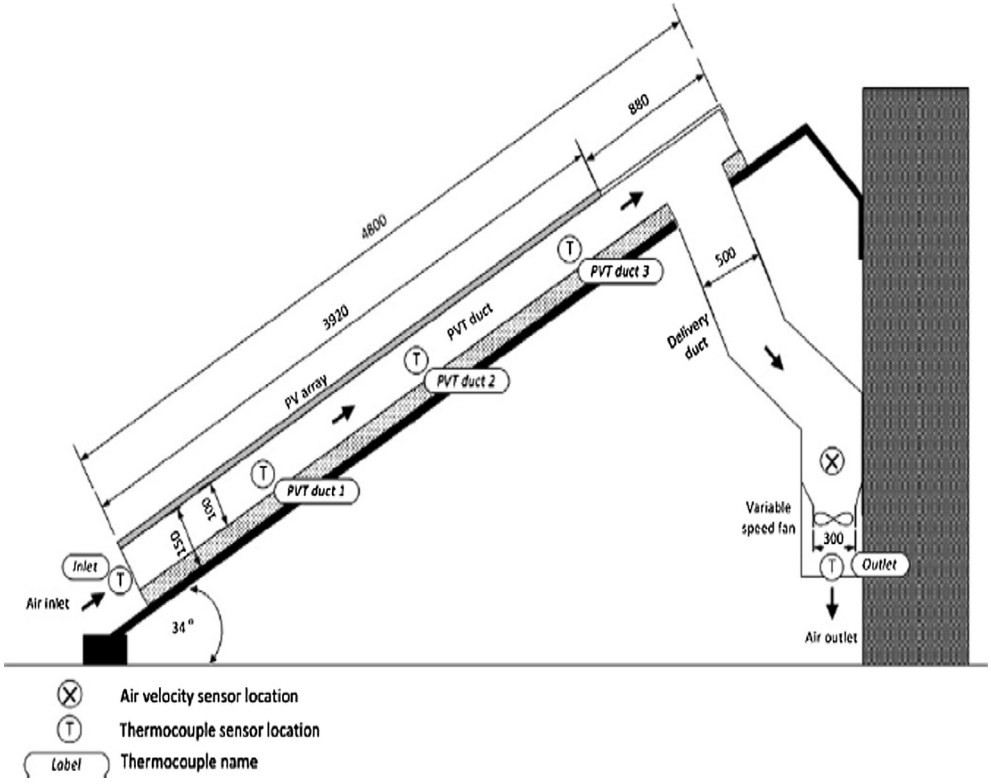


Fig. 2. PVT system schematic (Bambrook and Sproul, 2012).

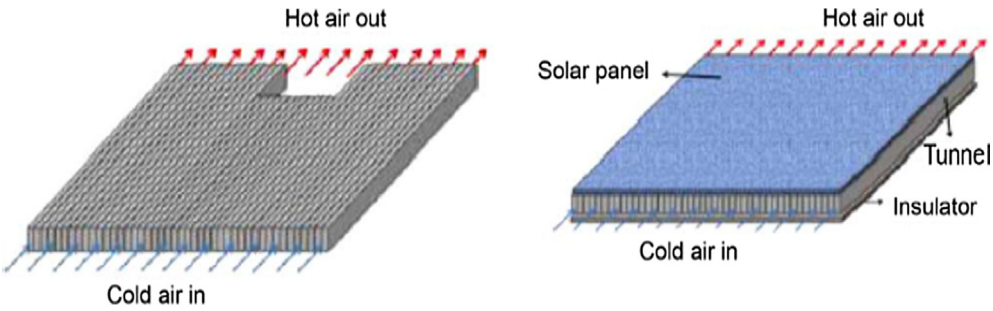


Fig. 3. Single pass PV/T with rectangular tunnel design (Jin et al., 2010).

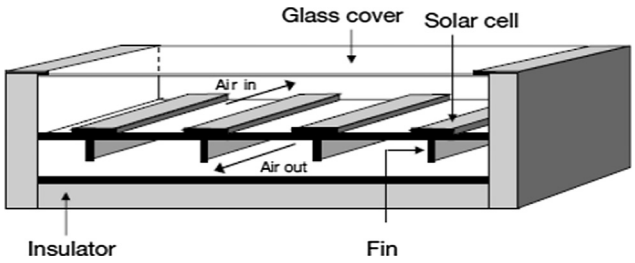


Fig. 4. The schematic model of a finned double-pass Photovoltaic/thermal air collector (Othman et al., 2007).

measurements. Also, it was noticed that the number of fans can play an important role in maximizing the electrical efficiency. On the other hand, the thermal efficiency can be increased be placing glass cover on the photovoltaic panels (see Fig. 6).

2.1.2. Concentrating PV/T air collector

Photovoltaic systems with concentrators are suitable for operation at high temperatures. A CPV/Thermal (CPV/T) system can be constructed by absorbing the rejected heat from concentrating photovoltaic systems (CPV), generating both heat and electricity (Tyagi et al., 2012). Recently, development of CPV/T collector is taking a considerable attention among researchers and scientists for more production of electricity and heat.

i. Performance an Design evaluation

The performance of double pass PV/T collector with Compound Parabolic Concentrator (CPC) and fins has been presented (Othman et al., 2005). As shown in Fig. 7, the absorber plate consists of array of photovoltaic cells, CPC and fins suspended at the back side of the absorber plate. A steady state model for predicting electrical and thermal performance was developed. Results proved that the electricity production in hybrid photovoltaic/thermal systems decreases as the air flow temperature increases.

Low concentrating PVT system collector and linear Fresnel concentrator which work on two-dimensional axis tracking system were constructed (Rosell et al., 2005). The overall combined efficiency was above 60% when the ratio of the concentration was above 6×.

Kostic et al. (2010a) constructed the optimal orientation of PV/T collector with reflectors. Flat reflectors were mounted on the hybrid

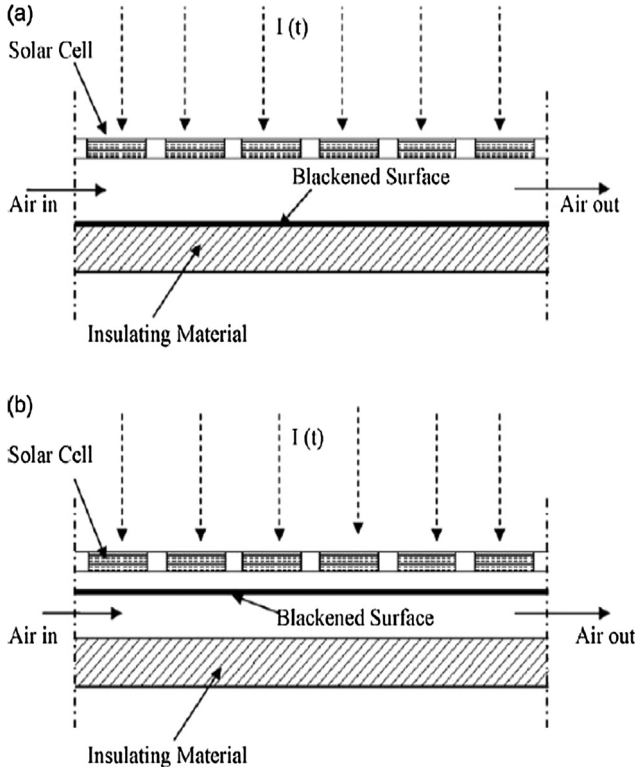


Fig. 5. (a) Cross-sectional view of air collector, air flows above the blackened surface. (b) Cross-sectional view of air collector, air flows below the blackened surface (Dubey et al., 2009).

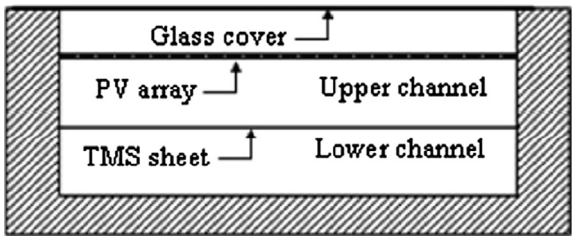


Fig. 6. Cross-sectional view of studied PV/T air collector with a thin aluminum sheet at the middle of the air channel (Shahsavar and Ameri, 2010).

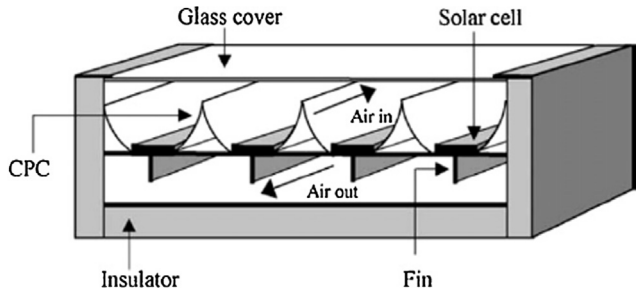


Fig. 7. The schematic model of a double-pass photovoltaic thermal solar collector with CPC and fins (Othman et al., 2005).

PV/T collector to improve the electrical and thermal energies. Besides, reflectors with movable PV/T collector system have been constructed to get a maximum solar radiation intensity. The total efficiency and energy-saving efficiency of PV/T collector evaluated by using the experimental data. Energy-saving efficiency for PV/T collector without reflectors and with reflectors was 60.1% and 46.7%, respectively.

Reflectance influence from flat aluminum concentrators on energy efficiency of PV/T collector was studied (Kostic et al., 2010b). It was noticed that Al foil results in maximizing the solar radiation intensity concentration factor.

Performance study of solar cell arrays based on a Trough Concentrating Photovoltaic/Thermal system (TCPV/T) was presented by using both theoretical and experimental analysis (Li, 2011). The optimum CR ratio for both single crystalline silicon cells and super cells were 4.23 and 8.46, respectively. On the other hand, investigation and optimization TCPV/T were carried out (Li et al., 2011). The overall performance of the proposed TCPV/T with Super cell arrays was evaluated by experimental measurements. Results showed that the concentrating silicon array is suitable for the proposed system with low and middle CR ratio.

2.2. PV/T liquid type collector

Liquid can be used as heat removal fluid in PV/T collector. Liquid type collector is more effective and practical than air type collector that is because of the temperature variation in liquid are much less compared to the air that is based on fluctuation on solar intensity levels.

2.2.1. Flat plate water collector

i. Performance and Design evaluation

Patrick et al. (2011) has studied the thermal and electrical performance of several single glazed flat plate PV/T concepts based on water circulation by using a simple two-dimensional thermal model under steady state conditions. The collector was tested by conducting an experiment using the single package lamination method, focusing on an improved heat transfer between PV cells and cooling fluid and on improved optical performance. The experimental results showed significant improvements in both thermal and electrical efficiencies of almost 88%.

Studies on the exergy and energy analysis of photovoltaic/thermal (PV/T) water collector with and without glass cover have been carried out (Chow et al., 2009). They found that the glazed PV/T collector is suitable for increasing the ambient temperature and the solar radiation. On the other hand, an unglazed collector is more favorable for increasing of the PV cell efficiency, wind velocity, water mass flow to collector area ratio and the packing factor.

Dubey and Tiwari (2009) performed analysis on PV/T water collector connected in series. By using theoretical modeling, the performance study of partially covered PV/T water collector connected in series was evaluated. Besides, they have developed an analytical expression for N number of collectors based on energy balance equations. They have presented an analysis of electrical energy, thermal energy and exergy by adjusting the numbers of collectors connected in series. Results showed that partially covered and fully covered PV/T water collector is suitable for hot water production and electrical generation, respectively.

Exergy analysis of integrated photovoltaic/thermal (IPVS) water heater (Fig. 8) under constant flow rate and constant collection temperature modes was studied (Tiwari et al., 2009). It was noticed that the maximum overall thermal and exergy efficiency was at 0.006 kg/s of hot water flow rate.

A Photovoltaic/Thermal Collector (PV/T) system with a new configuration has been proposed (Sakhr et al., 2013; Sakhr and Saqaff, 2014). The temperature of the outlet water, thermal, electrical and combined efficiencies of the PV/T system under different climatic conditions of Malaysia (sunny and cloudy) days have been studied. Based on the analytical results, the highest hourly thermal efficiency of sunny day is 61.3%, while it is a 59.6% for the cloudy day. The electrical efficiencies are 12.89% and 13.03% for typical sunny and cloudy, respectively. On the other hand, the highest hourly combined (PV/T) efficiencies are 74.1% and 72.63% for typical sunny and cloudy days, respectively.

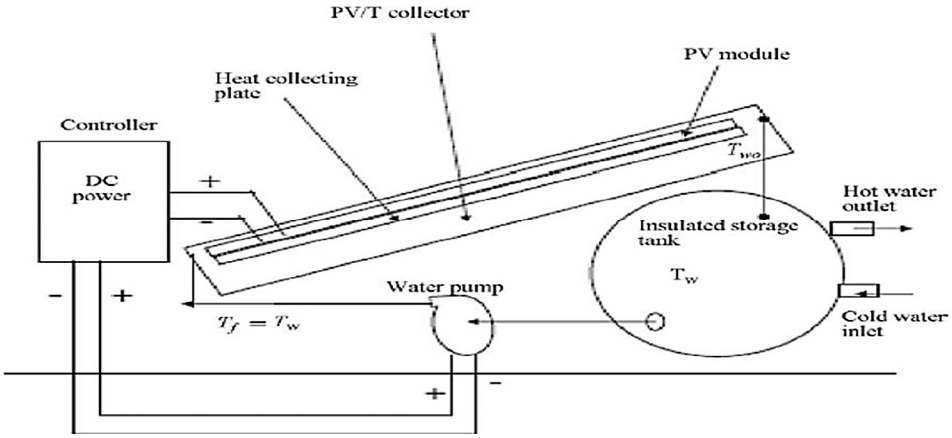


Fig. 8. Experimental setup of an integrated PV/T system (IPVTS) (Tiwari et al., 2009).

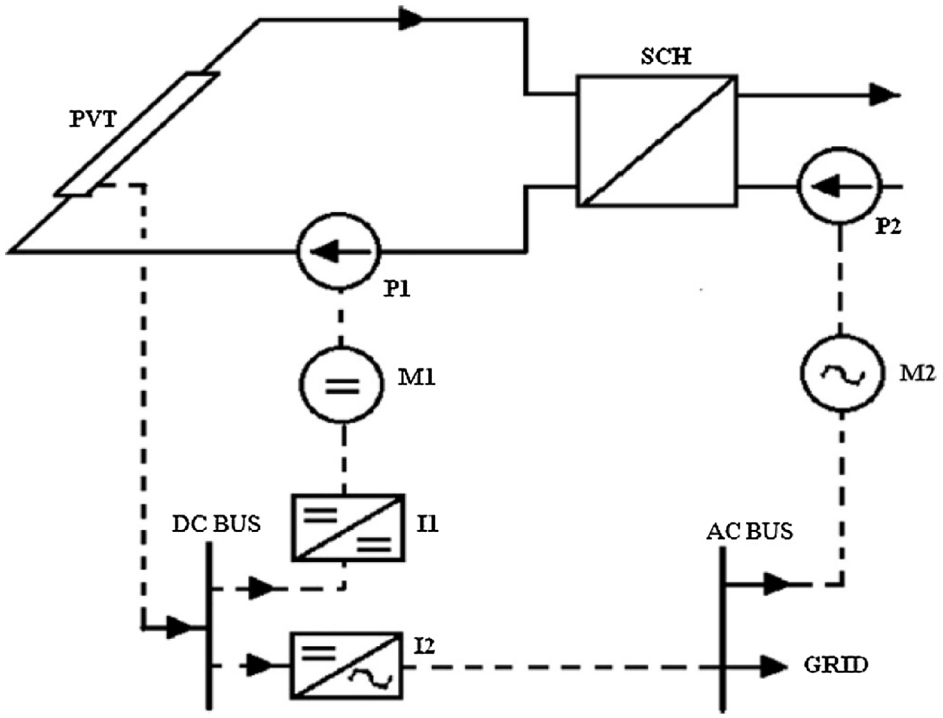


Fig. 9. Schematic of PVT system – Photovoltaic thermal system; SCH – heat exchanger; P1, P2 – pumps; M1-DC engine M1 – AC engine; I1 – DC controller, I2 – DC/AC inverter; DC – DC connection box; AC – AC connection box (Hazi et al., 2014).

Economical and technical opportunities assessment to use PVT systems in industry was studied (Hazi et al., 2014). Based on three parameters that are solar radiation, air and water supply temperature, an analytical model is developed to evaluate economic indicators and energy parameters. Therefore, the developed mathematical model is used to simulate the PVT system under different Romanian climatic locations. It was concluded that the water heating from the PVT system in industrial field is economically attractive and the payback period is lower than its lifetime (Fig. 9).

A water based PVT with a transient model was presented (Bilbao and Sproul, 2015). Based on the electrical-thermal analogy, a nodal approach was used for the new model to analyze the thermal response of the water based PVT system (Fig. 10). A one dimensional heat flow is modelled and average temperatures are used for the fluid and absorber plate to evaluate the PVT transient thermal output. Results reveal that the proposed new model was in a good agreement with the experimental measurement, and it could be used for product design and performance analysis.

2.2.1.1. Experimental work. The influence of different high sun radiation levels and fluid flow rate are examined in terms of outlet water and PV cell temperatures, electrical and thermal efficiencies and PVT overall system performance (Nasrin et al., 2018). To solve the three dimensional model numerical analysis, a finite element based software COMSOL Multiphysics has been used (Fig. 11). The analytical model results has been validated with available experimental measurements. It is observed that the PVT system overall performance increases with the increasing of the fluid flow rate and with an optimum flow rate of about 180 L/h. For increasing of sun radiations from 1000 to 5000 W/m^2, the electrical and thermal energy increase from 197 to 983 W to 1165–5387 W respectively.

PVT system with a novel design of thermal collector has been introduced (Nahar et al., 2017). A thermal collector with a parallel plate has been attached to the back surface of the photovoltaic module by means of thermal grease only, and the PVT system performance is evaluated analytically and validated by experimental measurements for various operating conditions (Fig. 12). A finite element method (FEM)

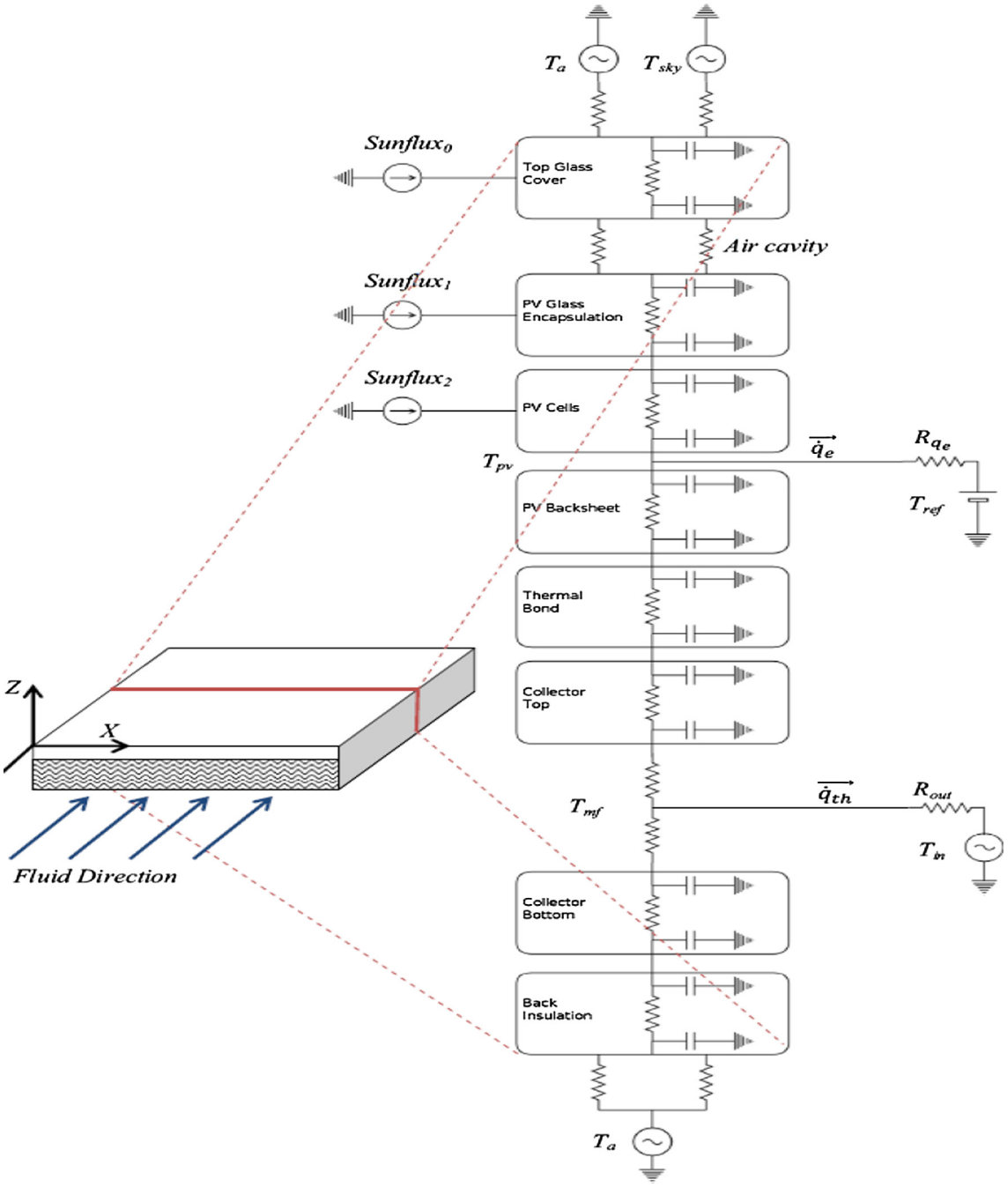


Fig. 10. Schematic for the RC model for glazed PVT (Bilbao and Sproul, 2015).

based software COMSOL Multiphysics was used to perform a 3D numerical analysis of the PVT system. Therefore, the outdoor experimental investigations has been carried out under the typical Malaysian climatic conditions. As a result, the analytical data are seen to be in a perfect agreement with the measurements taken from experiments. It was observed that the thermal performance of PVT system without and without absorber is the same. The numerical and experimental values for maximum overall PVT system efficiency was found to be 84.4% and 80%, respectively under the sun radiation of 1000 W/m^2 and inlet and ambient temperatures of 34 Â°C.

A retrofitted PVT system was assembled and installed together with a traditional PV panel (Sakellariou and Axaopoulos, 2017). An outdoor experiments were conducted to study the PVT and the PV performances. Results show that the PVT system can increase its average electrical efficiency by 0.32% and gain an additional thermal energy with average efficiency of 20.33%. Moreover, PVT's moderate thermal insulation revealed a detrimental factor regarding its thermal energy generation. The simulation output showed a good agreement with the experimental measurements, with average efficiencies of 3.12% for useful energy gain and between 4.46% and 10.72% for the average electrical output.

The connection mode of PVT systems and tilt angle are important factors that can influence the energy output of PVT systems (Sun et al., 2016). To demonstrate their effect, a water based PVT system which uses the thermosiphon phenomenon was constructed. The PVT systems are mounted vertically on residential buildings' facades (Fig. 13). Development of a dynamic model for the prediction of the water based PVT system was carried out. The analytical results were in good agreement with indoor experimental measurements. As compared with the parallel connection mode, electrical efficiency decreases by 2.0%

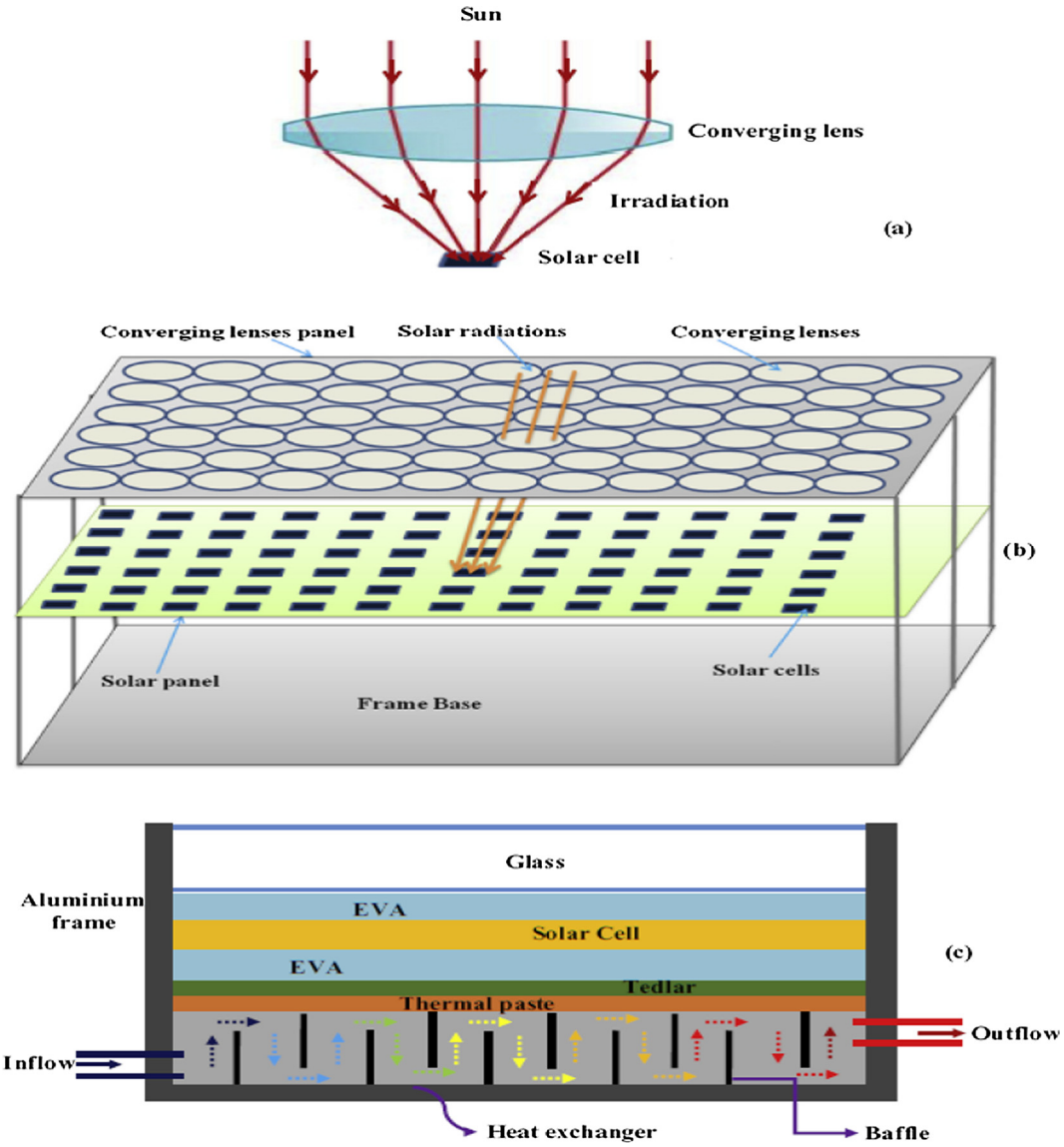


Fig. 11. Schematic diagram of (a) Fresnel lens with cell, (b) lens panel with solar panel and (c) PVT system (cross sectional view up to PV layers and top view in fluid flow layer) (Nasrin et al., 2018).

and it is increased by 11.4% for thermal energy and total energy increases by 5.4% for series connection mode. It was shown that the PVT system connection mode has a significant effect on the thermal and the electrical energy outputs. Therefore, a tilt angle of 20Â° of the PVT system can yield high energy benefits.

The mathematical model is presented for the PVT system (Khelifa et al., 2016). The PVT system consists of a photovoltaic panel and a thermal system for electricity production and heating the water. The thermal system consist of an absorber plate type sheet and tube that is placed under the PV surface for heat extraction, and hence cooling of the PV cells takes place, as a result, electrical efficiency can be increased (Fig. 14). The mathematical model was built based on the energy balance equations. The various layers and coolant temperatures of PVT system were predicted. The heat transfer and the fluid flow in the PVT system are studied using the ANSYS14 software. The effect of the transfer of heat phenomenon between the PV cells and the water is modeled. Therefore, the influence of sun radiation are taken into account when estimating the conditions for heat flux limit in the PVT layers.

The presentation of a mathematical model for PVT systems performance simulation, which takes into consideration all parameters and factors involved in the performance of an uncovered PVT system was carried out (Aste et al., 2015). The experimental data and the proposed model validation was performed in outdoor testing conditions on a traditional PVT product, at the Test Facility of the Politecnico di Milano University, Italy (Fig. 15).

The effects of various operating parameters, such as PV cell temperature, sin radiation, water mass flow rate, humidity, and dust, on the performance of a photovoltaic module are examined under outdoor operating conditions (Rahman et al., 2017). A tube with fins is attached under the PV module surface for cooling (Fig. 16). Results show that the PV efficiency of decreases by 5.82% as the temperature of the PV cell increases by 26.10 Â°C. Therefore, the PV efficiency also decreases by approximately 0.22% as the PV module temperature increases by 1 Â°C. As the solar radiation increases by 100 W/m^2, the temperature of the solar cell increases by 3.82 Â°C and the output power increases by 3.14 W. In conclusion, temperature of solar cell, sun radiation, water mass flow rate, humidity, and dust considerably affect the PV module performance.

An experimental investigation has been conducted to evaluate the series connected photovoltaic thermal (PVT) water collector performance in terms of overall thermal energy and exergy gain (Shyam et al.,

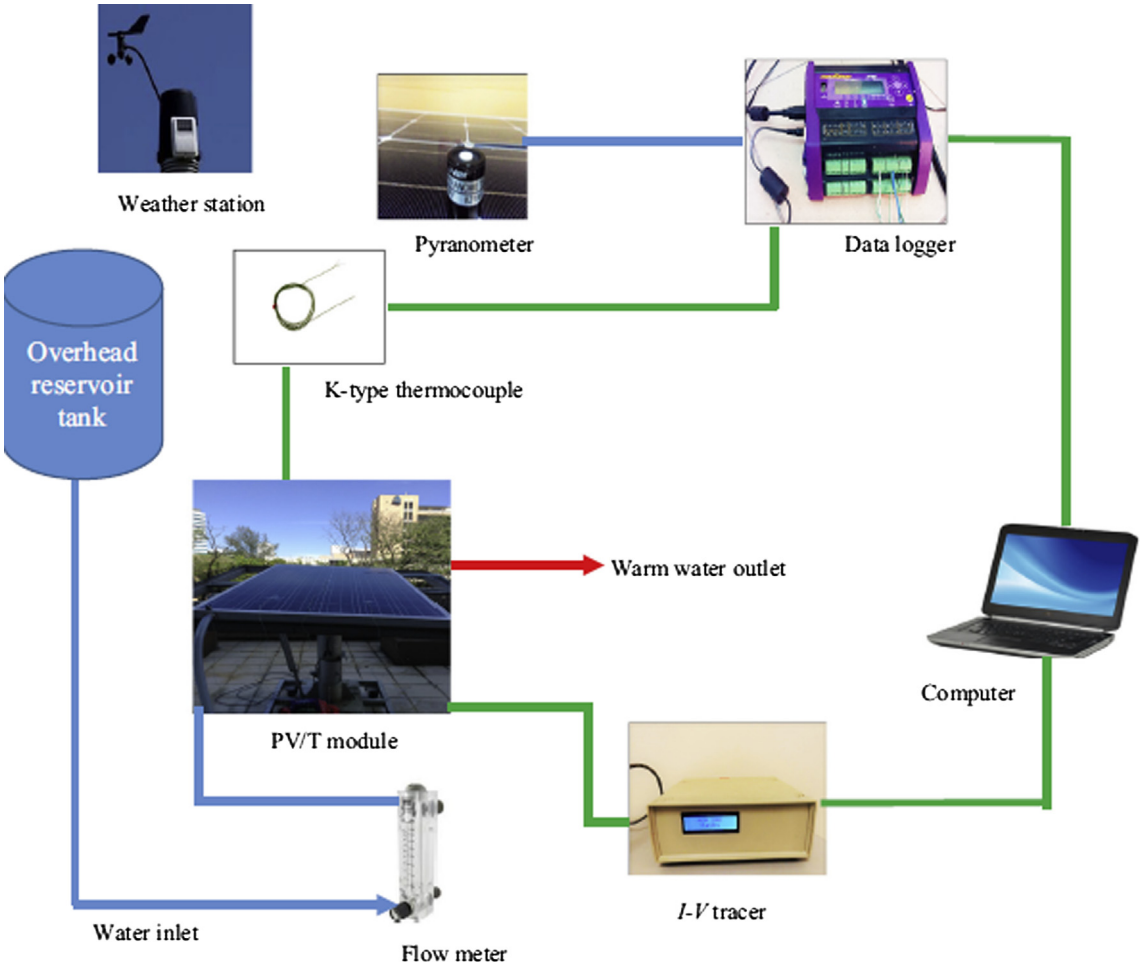


Fig. 12. Instrumentation of the experimental set up (Nahar et al., 2017).

2016). The analytical model has been validated by experiments in a clear sky, hazy and partly cloudy day for the climatic condition of New Delhi (Fig. 17). It was seen that analytical results were in good agreement with experimental measurements. The annual energy, exergy gain and CO_2 mitigation have also been elaborated.

Corry de Keizer (de Keizera and de Jonga, 2016) described the

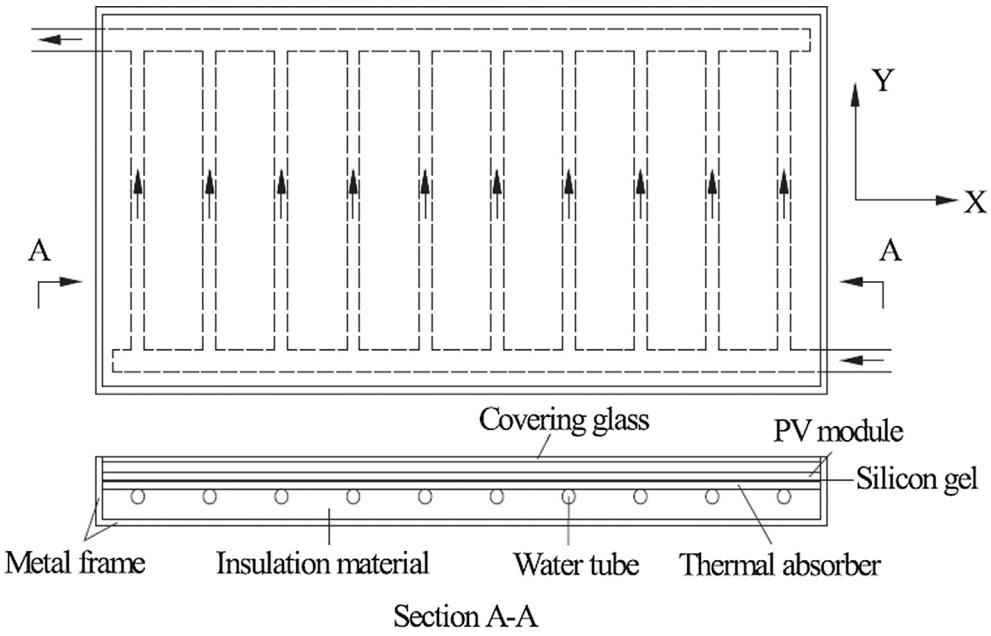


Fig. 13. Front view and cross section of PVT module Instrumentation (Sun et al., 2016).



Fig. 14. The heat exchanger bonded to the rear face of the PV module (Khelifa et al., 2016).

experimental outdoor field test facility used for measuring the electrical and thermal performance of the PVT system. Furthermore, the results of the steady state analysis and typical collector curves are presented. Therefore, results of this field test are used for dynamic simulations of the annual electrical and thermal energy yields for typical system designs and heat loads in the Netherlands. The performance of three different PVT collectors was evaluated. Within the WenSDak project, options for improving the performance are analyzed. Some measures, like applying a heat conducting paste, come at a certain cost that may not lead to the most cost-effective system solution.

Lämmle and Chirstoph (2016) presented a PVT collector concept with variable film insulation as overheating protection. An inflatable glass-film cushion regulates thermal losses. Performance and stagnation tests were carried out with a prototype. During normal operation the collector achieves a high thermal efficiency. In periods of standstill, the deflated cushion has a high heat dissipation rate by deactivating the low coating. Stagnation temperatures are thus limited to 95 °C. To conclude, the PVT collector combines the advantages of glazed and

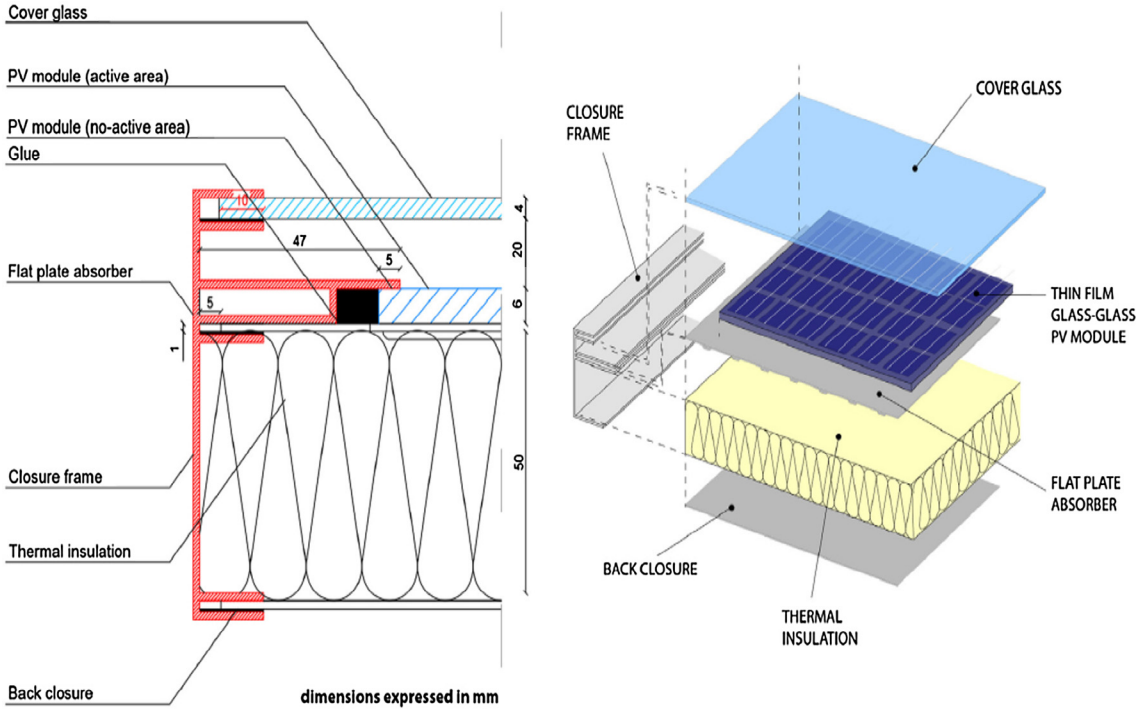


Fig. 15. Section and isometric view of the stratification of the collector. (Aste et al., 2015).



Fig. 16. PV module heat exchanger device (Rahman et al., 2017).

unglazed PVT collectors which are a high thermal efficiency and low stagnation temperatures.

The design of an innovative glazed water based PVT system and the development of a detailed mathematical model for the prediction of its electrical and thermal generation was carried out (Aste et al., 2016). The work presents a covered PVT collector design, utilizing thin film PV module and a flat plate with roll-bond absorber (Fig. 18). Finally, considerations about daily and annual energy yield of the proposed PVT collector as compared to a standard PV module was also elaborated. They concluded that more work must be concentrated on the performance of PVT collectors in summer in order to understand the regeneration effect of a-Si/lc-Si due to high temperatures.

Theoretical and experimental studies were conducted on a new configuration of the PVT system which extracts heat from the photovoltaic module (Touafek and Kerrour, 2014). This configuration is tube and sheet integrated into a prototype and tested at the unit of applied research in renewable energy in the south of Algeria. The advantages of this hybrid collector are better heat absorption and lower production cost compared to other configurations of hybrids collectors.

A photovoltaic thermal (PVT) collector performance was presented (Shan et al., 2013). PVT performance parameters such as the solar cell and the outlet water temperatures, as well as heat transfer rate and the photovoltaic efficiency can be determined using energy balance equations. The impacts of the series connected PVT module number, the mass flow rate and the inlet water temperature on the photovoltaic and thermal performances were elaborated (Fig. 19). The approaches of the thermal and electrical performances improvements were concluded. From results obtained, the less series connected PV modules, the lower inlet water temperature and the higher water mass flow rate will led to a high photovoltaic efficiency.

2.3. Concentrating PV/T water collector

i. Performance and Design evaluation

Mathematical and experimental investigations of five asymmetric PVT system with a flat plate receiver connected in series are studied (Fig. 20) (Koronaki and Nitsas, 2018). The performance evaluation in terms of energy and exergy thermal production under different operating conditions was discussed. The developed mathematical model combines optical, flow and thermal analysis for the first and second law efficiencies determination and it is validated with the experimental measurements with an adequate agreement being seen. Therefore, the performance of the PVT system is analyzed in terms of absorber temperature, thermal and exergy efficiency. After the thermal analysis, the mathematical model is further improved so as to take into consideration the electrical gain for the overall performance evaluation of the compound parabolic PVT solar collectors. Amongst the main findings, the experimental data proved that solar collectors that are connected in series work more efficiently throughout the year as they are able to harvest about 2.2 kW of useful energy gain in summer, 2.8 kW in spring

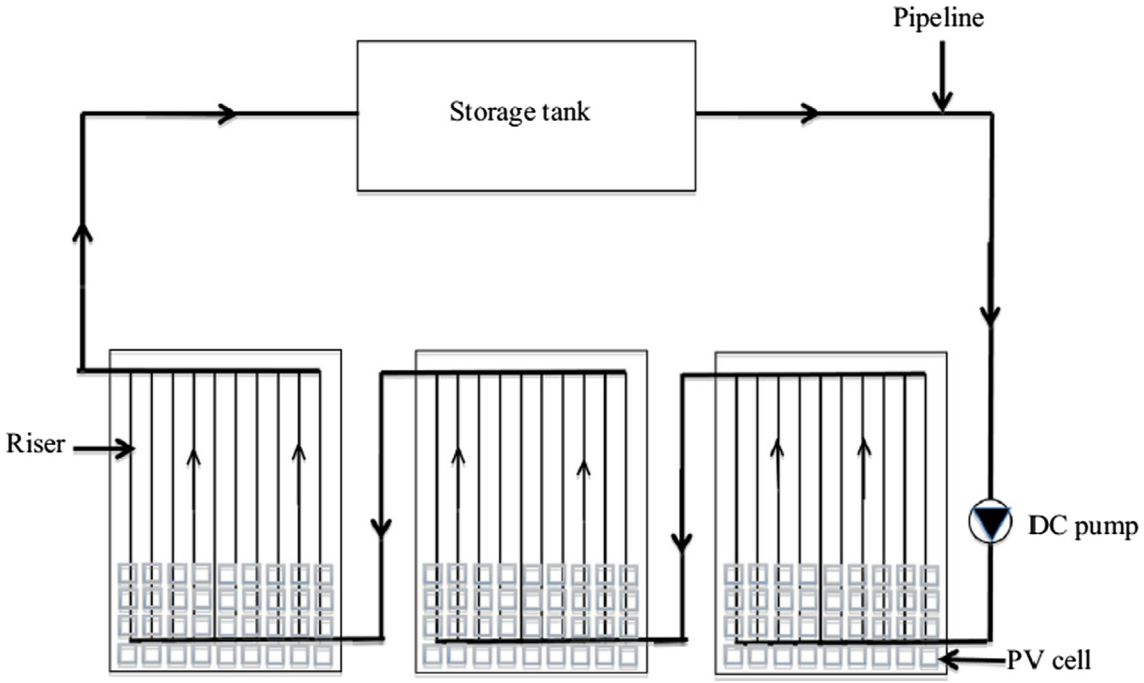


Fig. 17. Schematic and water flow pattern of the PVT water collector system (Shyam et al., 2016).

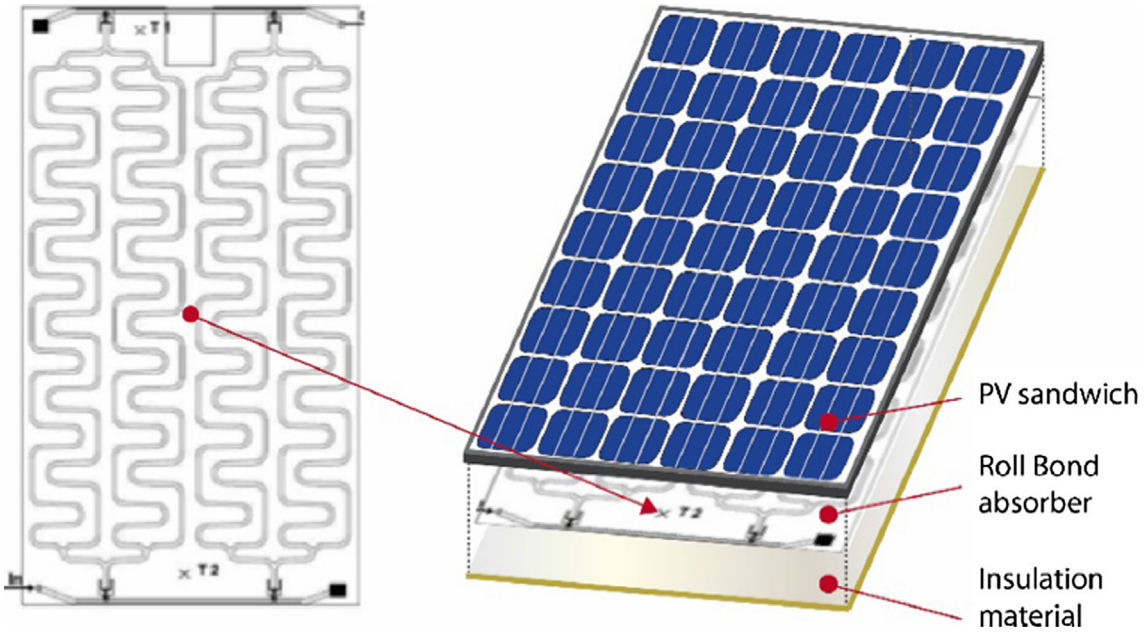


Fig. 18. Absorber plate and PVT collector configuration (Aste et al., 2016).

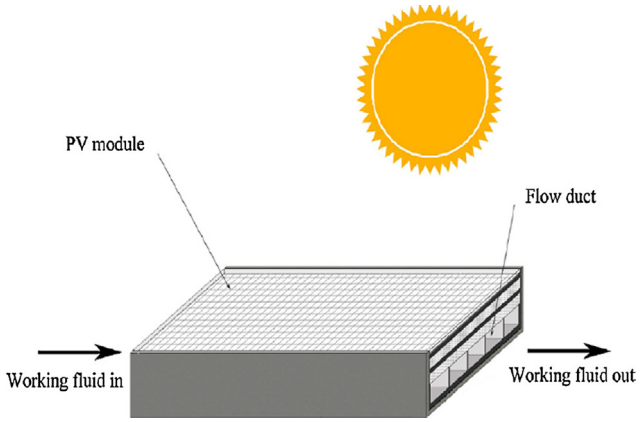


Fig. 19. Schematic diagram of a photovoltaic–thermal collector with water heating (Shan et al., 2013).



Fig. 20. The collector under study during the installation (Koronaki and Nitsas, 2018).

and 2.6 kW in autumn.

An experimental set up of fully covered concentrated photovoltaic thermal (CPVT) collector is designed and fabricated to enhance the system performance (Tripathi and Tiwari, 2017). This system is mounted at the building's rooftop of Centre for energy studies, IIT Delhi, India (Fig. 21). Therefore, a semitransparent photovoltaic panel is used to cover the CPVT system. A clear sky day condition for (September 2015 to August 2016) was taken into considerations. Two cases are studied on the basis of the rotation of the receiver, according to the movement of the sun, to analyze the annual behavior of present system, case (i): fixed position and case (ii): manual maximum power point tracking technique (M-MPPT). The manual tracking is adopted for 3 times a day in a 3 h interval (09.00–12.00–15.00 h). Consecutive days are chosen for both cases (i–ii) in each month throughout the year. It was observed that there is a good agreement between the theoretical analysis and the experimental data taken.

Coventry (2005) investigated the combined heat and power solar (CHAPS) photovoltaic thermal collector. Besides, a parabolic trough with concentration ratio of 37 was required for mono crystalline silicon cells and a two-axis tracking system. A tube with water and antifreeze were attached at the back of the cells to absorb most of the produced heat.

2.4. Combination of air/water type collector

2.4.1. Flat plate water collector

i. Performance and Design evaluation

Santbergen et al. (2010) carried out a detailed analysis on the energy yield of systems with covered sheet-and-tube PV/T collectors (Fig. 22). They showed that the combined annual electrical and thermal efficiencies of systems covered by sheet and tube PV/T collector are 15% lower than compared to separate PV and thermal collectors. These efficiencies can be improved by using the anti-reflective coating applications.

2.4.1.1. Experimental work. An experiment was conducted to evaluate the effect of mass flow rate on the thermal, electrical and overall combined efficiencies (Ibrahim et al., 2009). Besides, two different PV/T absorbers have been designed (Fig. 23). The first PV/T absorber called spiral flow absorber while second one known as single pass rectangular tunnel absorber. The result of the experiment showed that the combined PV/T efficiency of spiral flow absorber and the single pass rectangular tunnel absorber are 64% and 55%, respectively. They concluded that the spiral flow absorber gives a better performance

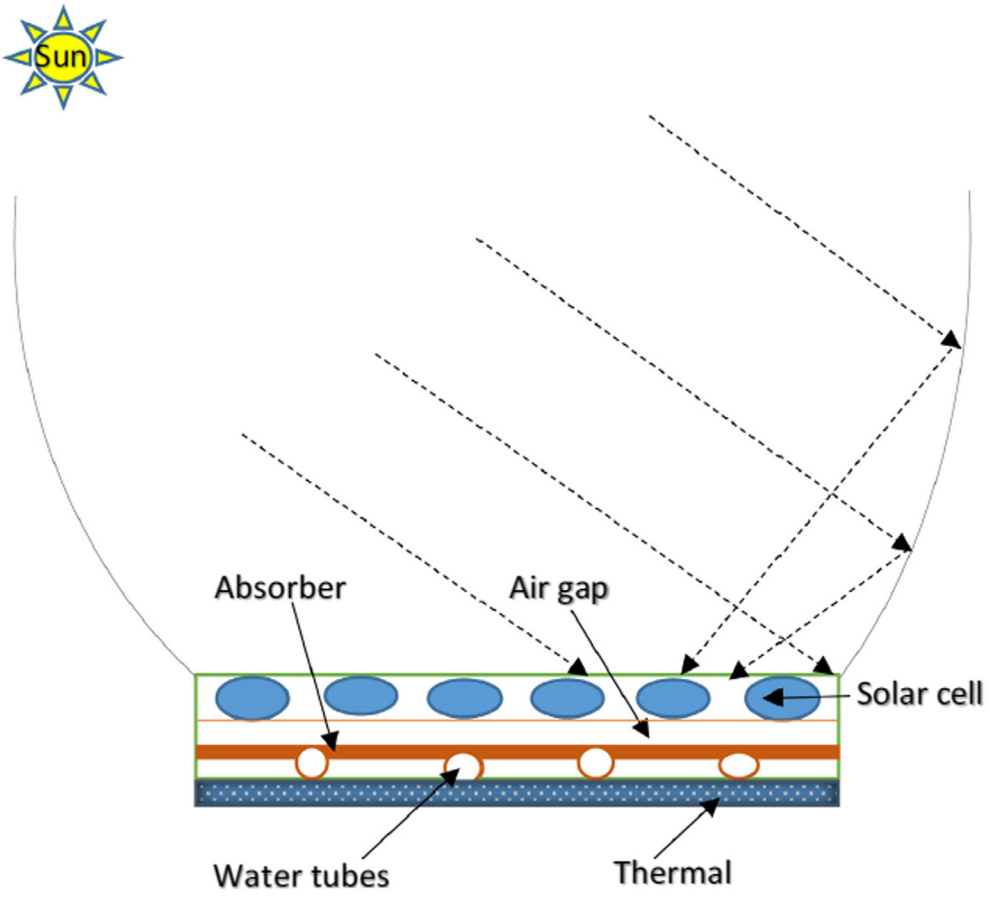


Fig. 21. Internal diagram for collector of fully covered photovoltaic thermal-compound parabolic concentrator (PVT-CPC) collector (Tripathi and Tiwari, 2017).

compared to single pass rectangular tunnel absorber.

2.5. PV/T applications

PV/T collector has various applications such as building integrated photovoltaic/thermal (BIPVT) collector, Photovoltaic-Thermal/heat pump systems, water desalination, solar still, solar cooling and solar greenhouse.

2.6. Photovoltaic-Thermal/heat pump system (PVT/heat pump)

Pei et al. (2008) presented a comparative study on the advantages of using a front cover. Results showed that the photothermic exergy

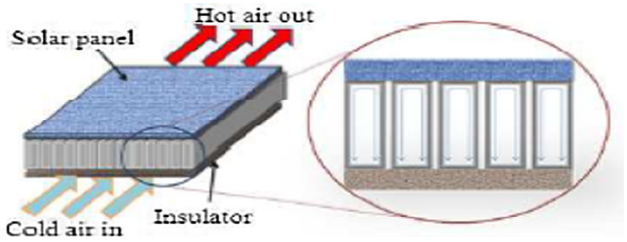


Fig. 23. The design of single pass rectangular tunnel absorber (Ibrahim and Jin, 2009).

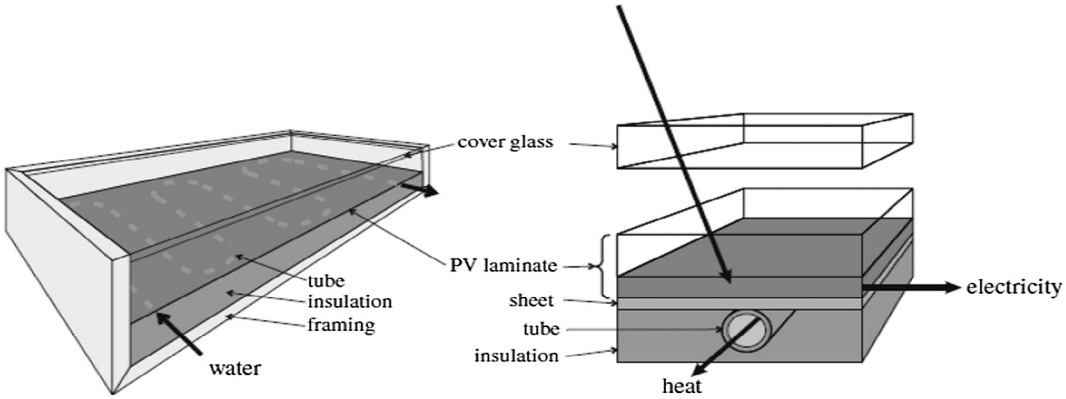


Fig. 22. A one-cover flat-plate sheet-and-tube PVT collector, converting incident solar irradiance into both electricity and heat. (Left) The complete collector. (Right) A detailed cross-section (Santbergen et al., 2010).

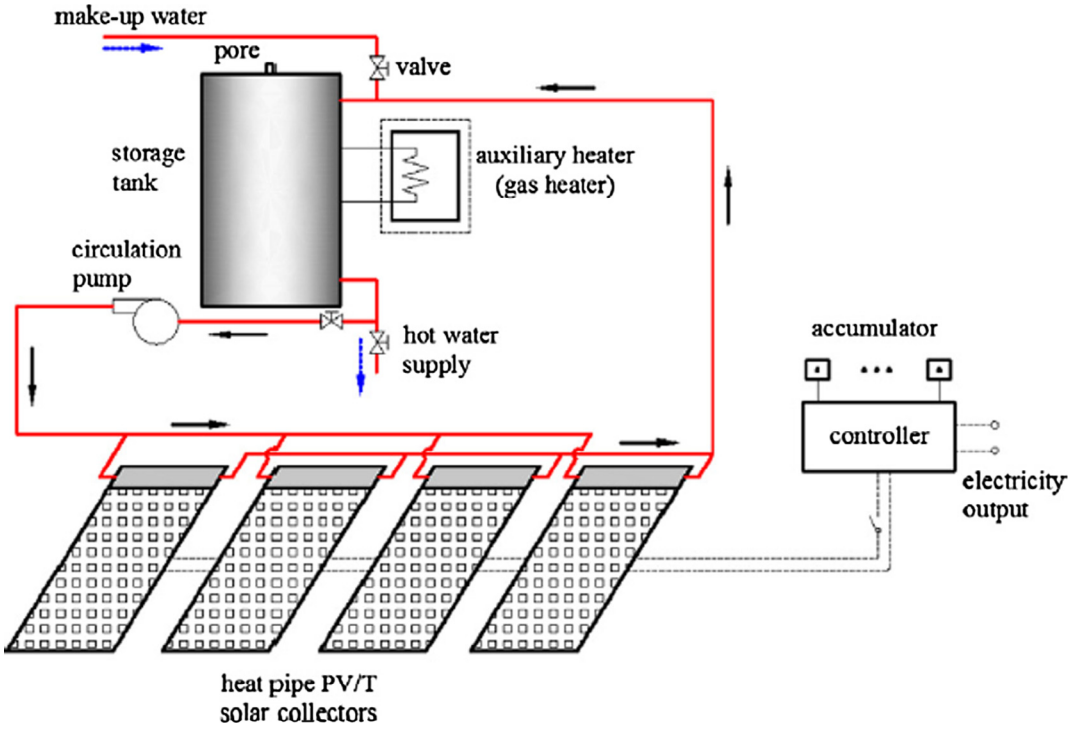


Fig. 24. Schematic diagram of the HP-PV/T system (Gang et al., 2012).

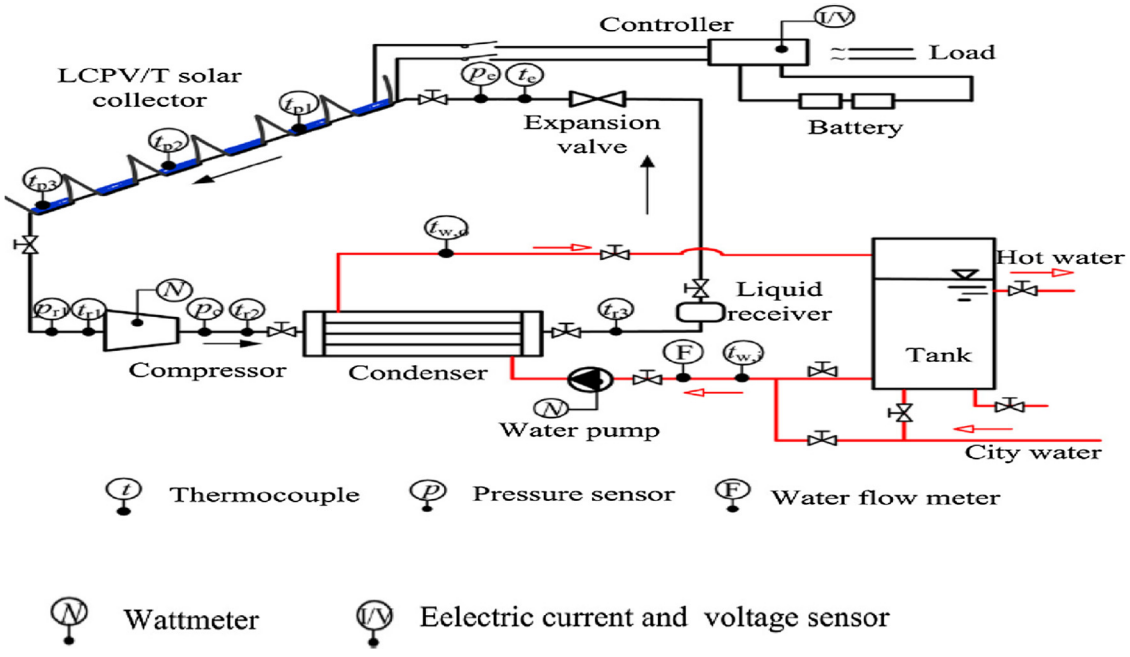


Fig. 25. Schematic diagram of the experimental prototype of the novel LCPV/T-HP system (Xu et al., 2011).

efficiency can be increased by single-glass cover, but has negative impacts on the photovoltaic exergy efficiency. Besides, the overall Exergy efficiency and the COP can be enhanced in the presence of the glass cover.

Chen and Wei (2011) performed a theoretical analysis on a novel Photovoltaic/Thermal Heat Pump system (PVT/Heat Pump). They have also developed a mathematical model to elaborate the heat pump energy performance as well as the thermal and electrical performance of PV cells. Simulation results showed that the average electrical and thermal efficiencies per month are 0.155 and 0.752, respectively.

A PVT/Heat Pump dynamic model was also evaluated (Gang et al., 2012). An experiment was performed to validate the measurements with the simulation results. The performance of the proposed system was studied under different climate conditions such as heat pipes tube space, PV cell covering factor of the collector and water flow rates.

Gang et al. (2012) have recently performed a simulation model of heat pipe PV/T systems to predict the annual thermal and electrical production in three different cities in China (Fig. 24).

Again, Gang et al. (2011) structured and designed PVT/heat pipe system. Besides, a comparative study was conducted between the proposed system and the conventional PV/T water type system. Results showed that the proposed can effectively operate in cold regions

without freezing. They have also developed a dynamic model to predict the performance of the system. Simulation values and experimental measurements were compared and validated. The electrical and thermal efficiencies were 9.4% and 41.9%, respectively.

The performance evaluation of a heat pump PV/T collector was considered (Shuang-Ying et al., 2011). Besides, a parametric elaboration was performed by varying relevant parameters such as packing factor for PV cells, inlet water temperature and heat loss coefficient. A theoretical model was also developed. The overall electrical, thermal and exergy efficiencies were 8.45%, 63.65% and10.26%, respectively.

An experiment on the PV/T heat pump air conditioning system was conducted (Fang et al., 2010). The performance parameters such as condensation pressure, the evaporation pressure and the coefficient of performance (COP) of PV/T heat pump air conditioning system were investigated. The experimental results showed that the electrical efficiency is almost 10.4% and the COP is 2.88.

A novel low-concentrating photovoltaic/thermal heat pump system has been developed (Xu et al., 2011) as it's shown in Fig. 25. Experiments on the proposed system were conducted in Nanjing, China. Furthermore, experimental results indicated that the novel LCPV/T-HP system achieved an averaged COP of 4.8 for water temperature from 30 _C to 70 _C on a sunny day, with an output electrical efficiency of 17.5%, which means 1.36 times higher than that of the conventional low concentrating photovoltaic (LCPV) system.

2.7. Building Integrated Photovoltaic/Thermal (BIPVT)

A temperature dependent analytical model for electrical and thermal efficiency of polycrystalline (p-Si), mono-crystalline (m-Si), cadmium telluride thin film (CdTe), amorphous silicon thin film (a-Si) and copper indium gallium selenide (CIGS) PV modules is developed with and without surface water flow (Tomar et al., 2018), as it is shown in Fig. 26. The developed model assist in determining the temperature influence on their performance of BIPVT system. The analytical analysis results are experimentally validated in outdoor operating conditions. The electrical and thermal energy output are calculated for both low and high water mass flow rate. Therefore, the daily average electrical energy output of PV modules; m-Si, p-Si, a-Si, CdTe and CIGS with and without water flow are found to be 12.30%, 10.98%, 6.08%, 6.60% and 7.71%, and 11.41%, 10.30%, 5.86%, 6.26% and 6.99% respectively. With considering a constant ambient temperature, the variation of water mass flow rate is determined.

An experimental model of Building Integrated Photovoltaic/Thermal air collector BIPV/T was designed and fabricated on roof of Faculty of Mechanical Engineering in University of Aleppo, Syria (Sayegh et al., 2011). The electrical efficiency was depending on the air flow rate. Form the experimental measurements, it was showed that the proposed system will provide a yearly saving about: electrical energy saving 1579 S.P/year (520 kW/year) and thermal energy 6133 S.P/year (2496 kW/year).

Flow and heat transfer in a Photovoltaic/Thermal collector (PV/T) for building application is discussed (Zogou and Stapountzis, 2012). The transient outdoor measurements with device testing have been reported. The results indicated that the heat transfer characteristics of the back sheet and the flow rate selection are critical to the PV façade performance.

Annual performance of building integrated photovoltaic/water-heating system (BiPVW) for warm climate application was presented (Chow et al., 2009). An energy model and a computer simulation were developed. Results indicate that the overall electrical and thermal efficiencies are 9.39% and 37.5%, respectively.

Agrawal and Tiwari (2010a,b) analyzed energy and exergy of the building integrated photovoltaic/thermal BiPVT system. To determine exergy, energy and life cycle cost assessment of the BiPVT system, a thermo-dynamical model was developed. Results showed that the exergy and energy efficiencies of amorphous silicon BiPVT were 7.13% and 33.54%, respectively, under the climate condition of New Delhi. They have also developed one-dimensional transient model. Analysis on PV performance, net exergy and energy gain were also reported. Results indicated that the system that is connected in series gives a better performance for a constant air mass flow rate, while system that is connected in parallel will give an enhanced performance for a constant

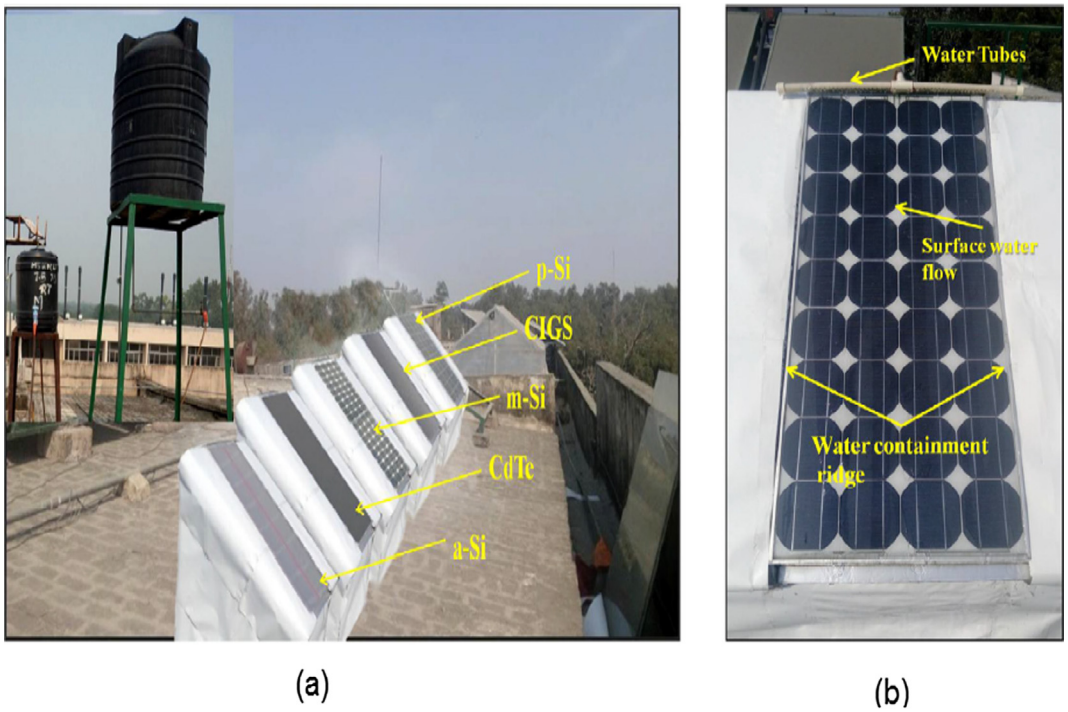


Fig. 26. Photograph views; (a) photovoltaic modules integrated on five identical test cells with water supply arrangement for continuous water flow, (b) close-view of surface water flow on module (Tomar et al., 2018).

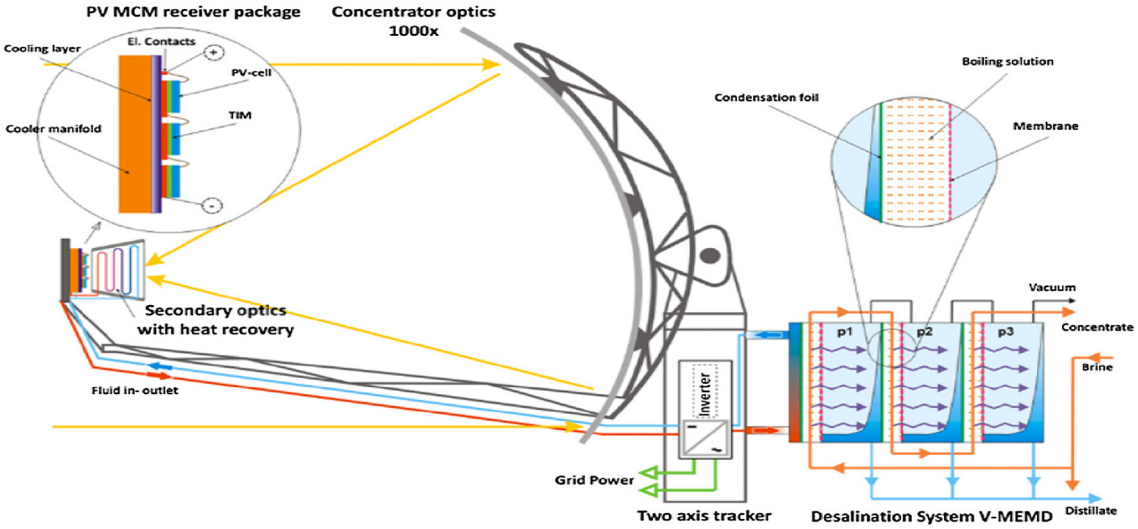


Fig. 27. The high concentration photovoltaic desalination (HCPVT) system with heat recovery (Koronaki and Nitsas, 2018).

air flow velocity.

2.8. Water desalination using PV/T collector

Mittelman et al. (2009) designed concentrating photovoltaic/thermal (CPVT) system with desalination. The coupled system generates electricity and simultaneously exploits the waste heat of the photovoltaic cells to desalinate water. A computer simulation was developed to calculate the annual production of water and electricity. Results showed that the combined system is a cost effective compared to other conventional solar desalination system.

The reuse of waste heat recovered from high concentration photovoltaic thermal (HCPVT) systems for saline and brackish water desalination is recently presented (Ong et al., 2012). They have developed a semi-empirical prediction model. This model is able to describe the multi effect membrane desalination MEMD system under steady state conditions. Experiments were conducted (Fig. 27) to validate the experimental measurements with the developed model. A ± 15% accuracy was gained as a result from the validation process.

3. Solar greenhouse using PV/T

Mittelman et al. (2007) performed exergy and energy analysis of PVT integrated with a solar greenhouse. Experiments were carried out to investigate the annual performance. Besides, numerical calculations were performed to validate the experimental results. It was noticed that the theoretical value of greenhouse air temperatures, tedlar back surface and solar cell was almost the same to the experimental results in terms of root mean square of percent deviation (7.05–17.58%) as well as correlation coefficient (0.95–0.97) and both show a reasonable agreement. Therefore, the exergy efficiency level was almost 4%.

3.1. Solar still

Dev and Tiwari (2010) have used two analytical methods to create the characteristics equation for photovoltaic/thermal active solar still. In the first method, linear and non-linear curves are plotted to find the instantaneous gain efficiency with respect to climatic and operational parameters. In the second method, the characteristics equations have been found using a matrix method. It has noticed that non-linear characteristics an equation is better that the linear one in terms of performance.

3.2. Solar air cooling

Sukamongkol et al. (2010) has examined the validity of a simulation model to predict the performance of a condenser heat recovery with a PV/T air collector to regenerate desiccant for reducing energy use of an air conditioning system. A reasonable agreement was observed between the experimental data and simulation results. The concluded that the use PV/T air collector in cooperation with condenser heat recovery can save the usage of energy of the air conditioning system by almost of 18%.

4. Conclusion

The combination of solar thermal and PV technologies leads to form a single module called PVT system. This system is able to produce heat and electricity simultaneously. Many hybrid PVT systems have been studied recently in terms of their overall performance. The current research in PVT technology focused on increasing the overall output energy and to find suitable solutions to the barriers that could limit the development of this technology in terms of design and performance considerations. This paper was reviewed the recent advances on PVT technology and applications, in term of performance, design, fabrication, simulation and experimental evaluations. Results of this paper showed that energy production of PVT systems has been increased significantly. As a general conclusion, PV/T system is a very promising device and needs a further development in term of the overall efficiency and cost reduction.

References

Agrawal, Basant, Tiwari, G.N., 2010a. Life cycle cost assessment of building integrated photovoltaic thermal (BIPVT) systems. Energy Build. 42, 1472–1481.
Agrawal, Basant, Tiwari, G.N., 2010b. Optimizing the energy and exergy of building integrated photovoltaic thermal (BIPVT) systems under cold climatic conditions. Appl. Energy 87, 417–426.
Alfegi, E.M.A., Sopian, K., Othman, M.Y.H., Yatim, B.B., 2009. The effect of flow rates on the performance of finned single pass, double duct photovoltaic thermal solar air heaters. Eur. J. Sci. Res. 25, 339–344.
Aste, Niccolò, Leonforte, Fabrizio, Del Pero, Claudio, 2015. Design, modeling and performance monitoring of a photovoltaic–thermal (PVT) water collector. Sol. Energy 112, 85–99.
Aste, Niccolò, Del Pero, Claudio, Leonforte, Fabrizio, Manfren, Massimiliano, 2016. Performance monitoring and modeling of an uncovered photovoltaic-thermal (PVT) water collector. Sol. Energy 135, 551–568.
Bambrook, S.M., Sproul, A.B., 2012. Maximising the energy output of a PVT air system. Sol. Energy 86 (6), 1857–1871.
Bilbao, J.I., Sproul, A.B., 2015. Detailed PVT-water model for transient analysis using RC networks. Sol. Energy 115, 680–693.

Chen, Hongbing, Wei, Ping, 2011. Numerical study on a novel photovoltaic/thermal heat pump system. Energy Proc. 12, 547–553.
Chow, T.T., Chan, A.L.S., Fong, K.F., Lin, Z., Heb, W., Ji, J., 2009. Annual performance of building-integrated photovoltaic/water-heating system for warm climate application. Appl. Energy 86, 689–696.
Chow, T.T., Pei, G., Fong, K.F., Lin, Z., Chan, A.L.S., Ji, J., 2009. Energy and exergy analysis of photovoltaic–thermal collector with and without glass cover. Appl. Energy 86, 310–316.
Coventry, J.S., 2005. Performance of a concentrating photovoltaic/thermal solar collector. Sol. Energy 78, 211–222.
de Keizera, Corry, de Jonga, Minne, 2016. Evaluating the thermal and electrical performance of several uncovered PVT collectors with a field test. Energy Proc. 91, 20–26.
Dev, Rahul, Tiwari, G.N., 2010. Characteristic equation of a hybrid (PV-T) active solar still. Desalination 254, 126–137.
Dubey, Swapnil, Tiwari, G.N., 2009. Analysis of PV/T flat plate water collectors connected in series. Sol. Energy 83, 1485–1498.
Dubey, Swapnil, Solanki, S.C., Tiwari, Arvind, 2009. Energy and exergy analysis of PV/T air collectors connected in series. Energy Build. 41 (863), 70.
Patrick, D., Christophe, M., Matthias, R., Hans-Martin, H., 2011. Efficient single glazed flat plate photovoltaic –thermal hybrid collector for domestic hot water system. Sol. Energy 85, 1457–1468.
Elhadidy, M.A., Shaahid, S.M., 2000. Parametric study of hybrid (wind + solar + diesel) power generating systems. Renew. Energy 21, 129–139.
Fang, Guiyin, Hu, Hainan, Xu, Liu, 2010. Experimental investigation on the photovoltaic–thermal solar heat pump air-conditioning system on water-heating mode. Exp. Therm. Fluid Sci. 34, 736–743.
Gang, Pei, Huide, Fu, Tao, Zhang, Jie, Ji, 2011. A numerical and experimental study on a heat pipe PV/T system. Sol. Energy 85, 911–921.
Gang, Pei, Huide, Fu, Huijuan, Zhu, Jie, Ji, 2012. Performance study and parametric analysis of a novel heat pipe PV/T system. Energy 37, 384–395.
Gang, Pei, Huide, Fu, Jie, Ji, Tin-tai, Chow, Tao, Zhang, 2012. Annual analysis of heat pipe PV/T systems for domestic hot water and electricity production. Energy Convers. Manage. 56, 8–21.
Hawlader, M.N.A., Jahangeer, K.A., 2006. Solar heat pump drying and water heating in the tropics. Sol. Energy 80, 492–499.
Hazi, Aneta, Hazi, Gheorghe, Grigore, Roxana, Vernica, Sorin, 2014. Opportunity to use PVT systems for water heating in industry. Appl. Therm. Eng. 63, 151–157.
Ibrahim, Adnan, Jin, Goh Li, Daghigh, Roonak, Salleh, Mohd Huzmin Mohamed, Othman, Mohd Yusof, Ruslan, Mohd Hafidz, Mat, Sohif, Sopian, Kamaruzzaman, 2009. Hybrid photovoltaic thermal (PV/T) air and water based solar collectors suitable for building integrated applications. Am. J. Environ. Sci. 5 (5), 618–624 (ISSN 1553-345).
Renewables in global energy supply: an IEA facts sheet, IEA/OECD, 2007.
Jin, G.L., Ibrahim, A., Chean, Y.K., Daghigh, R., Ruslan, H., Mat, S., et al., 2010. Evaluation of single-pass photovoltaic-thermal air collector with rectangle tunnel absorber. Am. J. Appl Sci. 7, 277–282.
Kamthania, Deepali, Nayak, Sujata, Tiwari, G.N., 2011. Performance evaluation of a hybrid photovoltaic thermal double pass facade for space heating. Energy Build. 43, 2274–2281.
Khelifa, A., Touafek, K., Ben Moussa, H., Tabet, I., 2016. Modeling and detailed study of hybrid photovoltaic thermal (PV/T) solar collector. Sol. Energy 135, 169–176.
Koronaki, I.P., Nitsas, M.T., 2018. Experimental and theoretical performance investigation of asymmetric photovoltaic/thermal hybrid solar collectors connected in series. Renew. Energy 118, 654–672.
Kostic, Lj T., Pavlovic, T.M., Pavlovic, Z.T., 2010a. Optimal design of orientation of PV/T collector with reflectors. Appl. Energy 87, 3023–3029.
Kostic, Ljiljana T., Pavlovic, Tomislav M., Pavlovic, Zoran T., 2010b. Influence of reflectance from flat aluminum concentrators on energy efficiency of PV/Thermal collector. Appl. Energy 87, 410–416.
Kumar, Rakesh, Rosen, Marc A., 2011. Performance evaluation of a double pass PV/T solar air heater with and without fins. Appl. Therm. Eng. 31, 1402–1410.
Lämmle, M., Chirstoph, B., 2016. A PVT collector concept with variable film insulation and low-emissivity coating. Energy Proc. 91, 72–77.
Li, Ming, Ji, Xu, Li, Guoliang, Wei, Shengxian, Li, YingFeng, Shi, Feng, 2011. Performance study of solar cell arrays based on a Trough Concentrating Photovoltaic/Thermal system. Appl. Energy 88, 3218–3227.
Li, M., Ji, X., Li, G.L., Yang, Z.M., Wei, S.X., Wang, L.L., 2011. Performance investigation and optimization of the Trough Concentrating Photovoltaic/Thermal system. Sol. Energy 85, 1028–1034.
Mittelman, Gur, Kribus, Abraham, Dayan, Abraham, 2007. Solar cooling with concentrating photovoltaic/thermal (CPVT) systems. Energy Convers. Manage. 48, 2481–2490.
Mittelman, Gur, Kribus, Abraham, Mouchtar, Ornit, Dayan, Abraham, 2009. Water desalination with concentrating photovoltaic/thermal (CPVT) systems. Sol. Energy 83, 1322–1334.
Nahar, Afroza, Hasanuzzaman, M., Rahim, N.A., 2017. Numerical and experimental investigation on the performance of a photovoltaic thermal collector with parallel plate flow channel under different operating conditions in Malaysia. Sol. Energy 144, 517–528.
Nasrin, R., Hasanuzzaman, M., Rahim, N.A., 2018. Effect of high irradiation and cooling on power, energy and performance of a PVT system. Renew. Energy 116, 552–569.
Ong, Chin Lee, Escher, W., Paredes, S., Khalil, A.S.G., Michel, B., 2012. A novel concept of energy reuse from high concentration photovoltaic thermal (HCPVT) system for desalination. Desalination 295, 70–81.
Othman, Mohd. Yusof Hj., Yatim, Baharudin, Sopian, Kamaruzzaman, Bakar, Mohd. Nazari Abu, 2005. Performance analysis of a double-pass photovoltaic/thermal (PV/T) solar collector with CPC and fins. Renew. Energy 30, 2005–2017.
Othman, Mohd. Yusof, Yatim, Baharudin, Sopian, Kamaruzzaman, Bakar, Mohd. Nazari Abu, 2007. Performance studies on a finned double-pass photovoltaic-thermal (PV/T) solar collector. Desalination 209, 43–49.
Pei, G., Ji, J., Chow, T.T., He, H., Liu, K., Yi, H., 2008. Comparative analysis of winter performance of PV-SAHP system with and without glass cover. J. Power Energy, Proc. IMechE Part A 179–187.
Rahman, Mohammad Mafizur, Hasanuzzaman, Md, Rahim, Nasrudin Abd, 2017. Effects of operational conditions on the energy efficiency of photovoltaic modules operating in Malaysia. J. Clean. Prod. 143, 912–924.
Rosell, J.I., Vallverdu, X., Lechon, M.A., Ibanez, M., 2005. Design and simulation of a low concentrating photovoltaic/thermal system. Energy Convers. Manage. 46, 3034–3046.
Sakellariou, Evangelos, Axaopoulos, Petros, 2017. Simulation and experimental performance analysis of a modified PV panel to a PVT collector. Sol. Energy 155, 715–726.
Sakhr, M., Saqaff, A., 2014. Performance analysis of the photovoltaic/thermal solar collector for different malaysian condition. J. Appl. Mech. Mater. 522–527.
Sakhr, M., Ervina, E., Saqaff, A., 2013. Theoretical study of new configuration of PVT system design. J. Adv. Mater. Res. 681–687.
Santbergen, R., Rindt, C.C.M., Zondag, H.A., van Zolingen, R.J.Ch., 2010. Detailed analysis of the energy yield of systems with covered sheet-and-tube PVT collectors. Sol. Energy 84, 867–878.
Sarhaddi, F., Farahat, S., Ajam, H., Behzadmehr, A., 2010. Exergetic performance assessment of a solar photovoltaic thermal (PV/T) air collector. Energy Build. 42, 2184–2199.
Sarhaddi, F., Farahat, S., Ajam, H., Behzadmehr, A., Mahdavi Adeli, M., 2010. An improved thermal and electrical model for a solar photovoltaic thermal (PV/T) air collector. Appl. Energy 87, 2328–2339.
Sayegh, Marderos Ara, Khayata, Nabhan, Nahhas, Tamar, 2011. Design and experimental study for using PV/T collectors in the Faculty of Mechanical Engineering University of Aleppo. Energy Procedia 6, 21–28.
Shahsavar, A., Ameri, M., 2010. Experimental investigation and modeling of a direct-coupled PV/T air collector. Sol. Energy 84 (84), 1938–1958.
Shan, Feng, Cao, Lei, Fang, Guiyin, 2013. Dynamic performances modeling of a photovoltaic–thermal collector with water heating in buildings. Energy Build. 66, 485–494.
Shuang-Ying, Wu, Zhang, Qiao-Ling, Xiao, Lan, Guo, Feng-Hua, 2011. A heat pipe photovoltaic/thermal (PV/T) hybrid system and its performance evaluation. Energy Build. 43, 3558–3567.
Shyam, Tiwari, G.N., Fischer, Olivier, Mishra, R.K., Al-Helal, I.M., 2016. Performance evaluation of N-photovoltaic thermal (PVT) water collectors partially covered by photovoltaic module connected in series: an experimental study. Sol. Energy 134, 302–313.
Sukamongkol, Y., Chungpaibulpatana, S., Limmeechokchai, B., Sripadungtham, P., 2010. Condenser heat recovery with a PV/T air heating collector to regenerate desiccant for reducing energy use of an air conditioning room. Energy Build. 42, 315–325.
Sun, L.L., Li, M., Yuan, Y.P., Cao, X.L., Lei, B., Yu, N.Y., 2016. Effect of tilt angle and connection mode of PVT modules on the energy efficiency of a hot water system for high-rise residential buildings. Renew. Energy 93, 291e301.
Tiwari, A., Dubey, S., Sandhu, Sodha, M.S., Anwar, S.I., 2009. Exergy analysis of integrated photovoltaic thermal solar water heater under constant flow rate and constant collection temperature modes. Appl. Energy 86, 2592–2597.
Tiwari, Arvind, Sodha, M.S., 2007. Parametric study of various configurations of hybrid PV/thermal air collector: experimental validation of theoretical model. Sol. Energy Mater. Sol. Cells 91, 17–28.
Tomar, Vivek, Tiwari, G.N., Bhatti, T.S., Norton, Brian, 2018. Thermal modeling and experimental evaluation of five different photovoltaic modules integrated on prototype test cells with and without water flow. Energy Convers. Manage. 165, 219–235.
Tonui, J.K., Tripanagnostopoulos, Y., 2008. Performance improvement of PV/T solar collectors with natural air flow operation. Sol. Energy 82, 1–12.
Touafek, K., Kerrour, F., 2014. Model validation of an empirical photovoltaic thermal (PV/T) collector. Energy Proc. 74, 1090–1099.
Tripanagnostopoulos, Y., 2007. Aspects and improvements of hybrid photovoltaic/thermal solar energy systems. Sol. Energy 81, 1117–1131.
Tripathi, Rohit, Tiwari, G.N., 2017. Annual performance evaluation (energy and exergy) of fully covered concentrated photovoltaic thermal (PVT) water collector: an experimental validation. Sol. Energy 146, 180–190.
Tyagi, V.V., Kaushik, S.C., Tyagi, S.K., 2012. Advancement in solar photovoltaic/thermal (PV/T) hybrid collector technology. Renew. Sustain. Energy Rev. 16, 1383–1398.
Xu, Guoying, Zhang, Xiaosong, Deng, Shiming, 2011. Experimental study on the operating characteristics of a novel low-concentrating solar photovoltaic/thermal integrated heat pump water heating system. Appl. Therm. Eng. 31, 3689–3695.
Zhang, Xingxing, Zhao, Xudong, Smith, Stefan, Xu, Jihuan, Yu, Xiaotong, 2012. Review of R&D progress and practical application of the solar photovoltaic/thermal (PV/T) technologies. Renew. Sustain. Energy Rev. 16, 599–617.
Zogou, Olympia, Stapountzis, Herricos, 2012. Flow and heat transfer inside a PV/T collector for building application. Appl. Energy 91, 103–115.